

1. **Experiment:** AlphaFold3-Based Protein Interface Analysis and His-Tag Purification

2. **Time:** 2026.05.21-2026.06.05

3. **Member:** Yang Jin, Qiwen Jiang, Yue Fu, Ziyang Yu, Zheyuan Deng, Ziqian He

4. **Method:**

(1) Candidate Protein Sequence Design

- ① For the functional screening of the target protein, 3 candidate protein sequences were designed via site-directed amino acid mutagenesis. Each sequence consists of three functional segments (A, B, and C) connected in tandem. Differentiated amino acid substitutions were introduced into different functional segments of the 3 candidate sequences to support subsequent screening of the optimal functional variant. The complete amino acid sequences of the 3 candidate proteins are listed below:

② Candidate Sequence 1:

- Segment A:

MVLQTQVFISLLWISGAYGIDPEKLKEEIKKLKEEKLEEFEPKYKELIKELTKKNLPKEELL
KEVEKFYKEFEKEFEKEIEKLKKDYKDDDDK

- Segment B:

APSAEQLAALAAEARAALLAQAAELRWTPFDLKLEVRPVPEGVLAIARATATRPQGPPVRL
RAEALLRLGGGSGGGSG

- Segment C:

FRAAVPLRIGDPVEIRLDDGTVPGRVDRVEGDVVSVALEPLTPGVHRVTVRFLTEPPQTSEPRL
VEFPPRVPIIVAVV

③ Candidate Sequence 2:

- Segment A:

MVLQTQVFISLLWISGAYGIDPEKLKEEIKKLKEEKLEEFEPKYKELIKELAKKNLPKEEV
MKEVEKFYKEFEKEFEKEIEKLKKDYKDDDDK

- Segment B:

APSAEQLAALAAEARAALLAQAAELGWTPFDLKLEVRPVPEGVLAIARATATRPQGPPVRL
RAEALLRLGGGSGGGSG

- Segment C:

FRAAVPLRVGNPVKIRLDDGTVPGRVDRVEGDVVSVALEPLTPGVHRVTVRFLTEPPQTS
EPRLVEFPPRVPIIVAVV

④ Candidate Sequence 3:

- Segment A:

MVLQTQVFISLLWISGAYGIDPEKLKEEIKKLKEEKLEEFEPKYKELIKELTKKNLPKEEVL
KEVEKFYKEFEKEFEKEIEKLKKDYKDDDDK

- Segment B:

APSAEQLAALAAEARAALLAQAAELRWTPFDLKLEVRPVPEGVLAIARATATRPQGPPVRL
RAEALLRLGGGSGGGSG

- Segment C:

FRAAVPLRVGDPVEIRMDDGTVPGRVDRVEGDVVSVALEPLTPGVHRVTVRFLTEPPQTS
EPRLVEFPPRVPIIVAVV

(2) AlphaFold3 structure prediction and interface analysis

- ① Sequence submission and prediction: The amino acid sequences of the three candidate proteins, sIL-6R α , and IL-6 were jointly submitted to the AlphaFold3 server (<https://alphafoldserver.com/>) as a single ternary complex. A total of 90 prediction runs were performed to ensure sufficient sampling of the conformational space. The 90 predicted models were ranked according to the interface predicted template modeling score (ipTM), and the top 5 models with the highest ipTM values were selected as

the high-quality model subset.

- ② Consensus model construction: An in-house Python 3.10 script (Biopython 1.87, NumPy 2.x) was used to average the coordinates of the 5 selected models. Using all C α atoms of sIL-6R α (chain A) as the reference set, the remaining models were superimposed onto the reference model one by one using the Kabsch least-squares superposition algorithm, and the superposition RMSD was recorded as a quality-control metric. The superimposed coordinates were then averaged point by point at the atomic level to obtain the consensus coordinates. The root mean square fluctuation (RMSF) of each atom was calculated and written into the B-factor column of the PDB file. The consensus model was examined for bond lengths, bond angles, and atom clashes, and exported in PDB format as the input structure for subsequent interface analysis.
- ③ Interface geometric parameter calculation: PyMOL was used for visualization and structural inspection of the consensus model. An in-house Python 3.10 script was used to calculate the buried surface area (BSA) of the interface and the hydrogen-bond interaction network. BSA was computed using the Shrake-Rupley algorithm (probe radius 1.4 Å, 960 sampling points per atom). The buried interface area was calculated as $\Delta\text{SASA} = \text{SASA}_{\text{chain1}} + \text{SASA}_{\text{chain2}} - \text{SASA}_{\text{complex}}$. Interchain hydrogen bonds were identified using a donor-acceptor distance cutoff of ≤ 3.5 Å and a D-H-A angle of $\geq 120^\circ$. The total number of hydrogen bonds was counted separately for the sIL-6R α (chain A)-candidate protein (chain C) and IL-6 (chain B)-candidate protein (chain C) interfaces.
- ④ Identification of interface residues: An in-house Python 3.12 script was used to parse the PDB coordinate file and compute the Euclidean distances between all interchain atom pairs to identify interface atom pairs and define interface residues. A distance cutoff of 6.0 Å was applied to the sIL-6R α (chain A)-candidate protein (chain C) interface. The IL-6 (chain B)-candidate protein (chain C) interface is relatively loose; after preliminary comparative tests, the cutoff for this interface was relaxed to 10.0 Å.
- ⑤ Weighted scoring of interface residues: Each interface atom pair was classified into an interaction type based on the chemical properties and atom names of the residues involved and the interatomic distance, covering salt bridges, hydrogen bonds, aromatic stacking, cation- π interactions, hydrophobic contacts, and van der Waals contacts. For each interface residue, the total number of interface contacts, the composition of interaction types, and the number of interacting partner residues were counted, and a hotspot score was calculated as follows:
$$\text{Score} = 0.5 \times \text{total contacts} + 2.0 \times \text{residue hotspot propensity} + 0.5 \times \text{number of partner residues} + \text{interaction weights}.$$

The interaction weights were assigned as follows: salt bridge, 5.0; hydrogen bond, aromatic stacking, and cation- π interaction, 3.0 each; hydrophobic contact, 1.0; van der Waals contact, 0.5. The residue hotspot propensities were assigned according to known statistical tendencies.
- ⑥ Hotspot residue determination: For each chain interface, the mean and standard deviation of the scores of all interface residues were calculated, and the scores were normalized as $Z = (\text{Score} - \mu)/\sigma$. Residues with $Z > 1.98$ were defined as hotspot residues. This scoring scheme simulates the principle of alanine scanning: residues with higher scores concentrate more interactions at the interface and are expected to be more disruptive to binding when mutated to alanine.
- ⑦ Mutation annotation and visualization: Hotspot residues were annotated as mutations using single-letter amino acid codes. The alanine scanning results of S1-S3 were visualized as bar plots using matplotlib, with the screened mutation sites on the x-axis and the interface hotspot scores on the y-axis, to compare the distribution and conservation of key interface residues among the three candidate proteins.

(3) Contracted Protein Expression and Purification

- ① The 3 candidate protein sequences above were formally submitted to Yaoqiang Company to perform the full core workflow of His-tagged protein expression and affinity purification. The company operates with standardized protocols, and the core procedures are as follows:
- ② Gene synthesis and vector construction: Codon optimization for *E. coli* expression preference was performed based on the input amino acid sequences. The target genes were chemically synthesized and cloned into the pET-28a expression vector to generate recombinant constructs carrying an N-terminal 6×His tag. Sequence accuracy was confirmed by Sanger sequencing.
- ③ Induced protein expression: The recombinant plasmids were transformed into *E. coli* BL21 competent cells. A single colony was inoculated into selective LB medium and cultured with shaking at 37 °C until OD600 reached 0.7. IPTG was added to induce protein expression under optimized conditions, and cell pellets were harvested by centrifugation.
- ④ Cell lysis and supernatant preparation: Cell pellets were resuspended in lysis buffer and disrupted by sonication on ice. Cellular debris was removed by centrifugation, and the supernatant containing soluble target proteins was collected for purification.
- ⑤ Ni-NTA affinity chromatography: Purification was performed using Ni-NTA agarose columns. The workflow included column equilibration, sample loading and binding, removal of nonspecific impurities with low-concentration imidazole, and competitive elution of target proteins with high-concentration imidazole. Fractions corresponding to the protein elution peak were collected.
- ⑥ Sample processing and delivery: The purified proteins were processed via dialysis buffer exchange, concentration calibration, sterile filtration and aliquoting, then delivered to the laboratory via cold chain transportation.

5. Result:

Table 1. 90 repeated AlphaFold3 predictions of the S1 virtual complex

	ipTM	pTM	BSA (Å²)	Hydrogen bond analysis		Hydrophobic pairing	
				effective bonds	average energy	mimic with IL6	effective bonds
S1_1	0.51	0.63	2349.764	18	-26.34	17	12
S1_2	0.51	0.63	1360.39	11	-16.81	15	13
S1_3	0.53	0.64	2024.243	9	-10.05	14	13
S1_4	0.51	0.63	2032.243	14	-16.34	16	14
S1_5	0.53	0.64	2063.58	11	-16.78	16	13
S1_6	0.51	0.63	1909.652	8	-11.11	16	11
S1_7	0.52	0.63	2143.567	17	-17.12	16	22
S1_8	0.51	0.63	1878.599	14	-18.28	16	26
S1_9	0.52	0.63	2120.488	11	-10.39	30	11
S1_10	0.52	0.63	2316.62	10	-13.42	17	11
S1_11	0.52	0.64	1914.565	13	-9.45	15	14
S1_12	0.51	0.63	1924.331	14	-21.45	15	10
S1_13	0.51	0.63	1893.104	15	-24.51	16	19
S1_14	0.52	0.63	2778.603	15	-18.76	31	14
S1_15	0.53	0.64	1744.917	15	-18.56	15	9
S1_16	0.51	0.63	1849.511	14	-14.88	16	9
S1_17	0.5	0.62	1835.762	11	-10.75	17	8
S1_18	0.53	0.64	1671.753	11	-17.97	15	9
S1_19	0.51	0.62	1706.979	8	-9.82	16	26
S1_20	0.5	0.62	1744.795	20	-18.83	15	23

S1_21	0.5	0.62	2085.944	12	-18.02	20	11
S1_22	0.52	0.63	1583.452	18	-20.56	17	26
S1_23	0.51	0.63	1841.118	13	-10.3	14	25
S1_24	0.52	0.64	1863.401	18	-22.93	23	26
S1_25	0.52	0.63	2123.418	14	-13.25	31	26
S1_26	0.51	0.63	1936.263	15	-22.12	30	14
S1_27	0.52	0.64	1967.832	17	-12.19	28	19
S1_28	0.52	0.64	2695.306	20	-18.94	25	25
S1_29	0.51	0.63	1369.399	15	-10.44	24	11
S1_30	0.51	0.63	1850.636	9	-15.19	19	19
S1_31	0.51	0.63	1617.287	18	-25.89	17	13
S1_32	0.52	0.63	2130.463	12	-24.04	23	13
S1_33	0.52	0.63	2457.421	15	-11.18	18	18
S1_34	0.51	0.63	1834.801	10	-21.48	24	17
S1_35	0.51	0.63	1593.516	8	-22.82	19	12
S1_36	0.51	0.63	2222.373	20	-13.94	19	18
S1_37	0.52	0.64	1885.885	9	-11.36	25	18
S1_38	0.51	0.63	1817.928	19	-24.61	22	16
S1_39	0.56	0.65	2146.613	15	-22.3	15	19
S1_40	0.52	0.64	2456.064	18	-22.61	31	16
S1_41	0.5	0.62	2234.849	13	-17.07	21	24
S1_42	0.51	0.63	1970.99	16	-14.64	16	12
S1_43	0.51	0.63	2047.242	15	-21.25	28	18
S1_44	0.51	0.63	1984.277	9	-25.95	31	8
S1_45	0.53	0.64	1912.466	19	-10.84	14	15
S1_46	0.53	0.63	1899.164	20	-17.35	22	21
S1_47	0.51	0.63	1851.335	16	-16.17	29	8
S1_48	0.5	0.64	1670.124	9	-12.36	31	23
S1_49	0.52	0.63	1421.924	15	-10.1	27	8
S1_50	0.51	0.63	1606.133	10	-12.08	25	17
S1_51	0.52	0.63	2238.229	14	-18.31	14	21
S1_52	0.52	0.64	1807.015	14	-15.07	29	21
S1_53	0.53	0.63	1683.641	13	-24.42	22	25
S1_54	0.52	0.64	1475.981	17	-17.48	16	9
S1_55	0.52	0.62	1651.728	19	-13.06	21	9
S1_56	0.51	0.64	2026.674	15	-13.5	20	26
S1_57	0.52	0.64	1872.049	8	-12.07	26	8
S1_58	0.5	0.64	1641.598	13	-15.32	31	24
S1_59	0.51	0.62	1411.808	11	-20.05	27	19
S1_60	0.52	0.64	1501.604	9	-25.99	21	19
S1_61	0.51	0.64	1567.534	14	-10.49	25	11
S1_62	0.51	0.62	1446.498	17	-11.63	23	23
S1_63	0.52	0.63	1564.898	10	-22.74	23	17
S1_64	0.53	0.63	2099.609	16	-25.91	27	19
S1_65	0.52	0.64	2005.774	16	-10.96	28	18
S1_66	0.5	0.62	2007.084	12	-22.56	30	9

S1_67	0.53	0.63	1616.755	17	-14.36	31	26
S1_68	0.52	0.63	1788.079	9	-24.02	28	8
S1_69	0.53	0.63	1434.374	20	-11.98	30	26
S1_70	0.51	0.64	1974.08	8	-17.15	19	20
S1_71	0.52	0.62	1810.24	20	-10.32	28	24
S1_72	0.52	0.63	1773.291	10	-25.81	25	13
S1_73	0.51	0.63	2034.549	9	-12.19	14	12
S1_74	0.52	0.63	1394.483	14	-14.88	17	19
S1_75	0.51	0.63	1521.69	20	-24.34	24	24
S1_76	0.5	0.63	1453.746	16	-20.79	29	22
S1_77	0.52	0.64	1860.97	16	-20.83	23	12
S1_78	0.51	0.62	1622.152	12	-21.13	18	9
S1_79	0.52	0.62	1465.983	10	-18.31	31	12
S1_80	0.5	0.63	1716.267	11	-17.67	31	24
S1_81	0.51	0.63	1557.368	15	-17.17	14	15
S1_82	0.52	0.64	1835.905	8	-21.39	22	22
S1_83	0.5	0.62	1568.222	10	-13.22	25	13
S1_84	0.51	0.62	1748.81	18	-11.15	27	21
S1_85	0.51	0.63	1654.912	19	-14.27	31	14
S1_86	0.51	0.63	1532.144	13	-16.06	25	8
S1_87	0.52	0.63	1636.28	18	-21.98	31	19
S1_88	0.51	0.62	1545.976	11	-19.53	15	20
S1_89	0.51	0.63	1701.117	19	-26.04	14	23
S1_90	0.51	0.63	1594.928	16	-23.28	28	8

A total of 90 independent AlphaFold3 predictions were performed shown. ipTM, interface predicted TM-score (0–1, higher indicates higher confidence). pTM, predicted TM-score. BSA ($\text{\AA}^2$), buried surface area in square angstroms. For Hydrogen bond analysis, effective bonds denote the number of interfacial hydrogen bonds, and average energy represents the average interaction energy (in kcal/mol). mimic with IL6 denotes the number of interfacial hydrogen bonds that recapitulate the native binding interfaces of IL-6. For Hydrophobic pairing, effective bonds indicate the number of interfacial hydrophobic contacts.

Table 2. 90 repeated AlphaFold3 predictions of the S2 virtual complex

	ipTM	pTM	BSA (Å²)	Hydrogen bond analysis		Hydrophobic pairing	
				effective bonds	average energy	mimic with IL6	effective bonds
S2_1	0.52	0.63	1695.32	13	-20.24	18	3
S2_2	0.56	0.65	1905.99	12	-15.5	18	9
S2_3	0.52	0.64	1845.903	15	-14.56	17	11
S2_4	0.54	0.65	1288.401	15	-16.36	17	16
S2_5	0.52	0.63	2077.535	9	-9.72	17	17
S2_6	0.57	0.64	1782.901	14	-17.4	18	9
S2_7	0.53	0.64	1653.865	7	-10.81	19	2
S2_8	0.59	0.65	2146.193	10	-17.02	19	20
S2_9	0.55	0.63	1327.702	10	-15.01	19	3
S2_10	0.55	0.64	1545.423	11	-15.9	19	7
S2_11	0.53	0.64	2075.655	11	-20.21	17	8
S2_12	0.68	0.67	1570.663	9	-14.09	17	17
S2_13	0.54	0.64	1666.834	9	-19.07	19	7
S2_14	0.61	0.65	1981.477	7	-9.96	19	12
S2_15	0.53	0.64	1713.967	12	-19.55	19	18
S2_16	0.56	0.65	1337.438	13	-12.84	18	3
S2_17	0.55	0.64	2051.938	14	-17.14	17	18
S2_18	0.57	0.64	2016.807	9	-13.48	17	12
S2_19	0.57	0.64	1393.555	12	-16.83	19	11
S2_20	0.54	0.63	1608.409	8	-17.25	19	13
S2_21	0.65	0.63	1862.986	9	-19.61	18	5
S2_22	0.69	0.65	1743.842	11	-14.24	19	2
S2_23	0.59	0.66	1748.187	14	-11.21	17	9
S2_24	0.64	0.63	1618.889	14	-12.62	17	2
S2_25	0.68	0.63	1877.445	15	-15.5	18	14
S2_26	0.54	0.63	1323.088	8	-15.42	17	13
S2_27	0.54	0.63	1608.409	8	-16.35	17	10
S2_28	0.56	0.66	1802.612	15	-19.13	17	14
S2_29	0.52	0.63	1646.468	12	-11.83	18	17
S2_30	0.62	0.64	1919.159	10	-13.91	19	15
S2_31	0.59	0.63	1988.027	12	-11.91	19	7
S2_32	0.65	0.64	1676.653	12	-10.02	18	12
S2_33	0.56	0.64	1959.989	9	-10.9	18	2
S2_34	0.53	0.63	1327.061	15	-12.66	18	9
S2_35	0.71	0.63	1860.51	9	-12.13	17	3
S2_36	0.56	0.66	1991.81	7	-14.04	17	19
S2_37	0.56	0.63	1494.077	14	-11.36	19	4
S2_38	0.56	0.66	1749.186	12	-16.52	18	20
S2_39	0.52	0.65	1803.109	9	-18.55	18	15
S2_40	0.6	0.64	1329.033	13	-10.81	18	3
S2_41	0.53	0.64	1849.439	10	-14.5	19	16
S2_42	0.71	0.67	1590.004	10	-13.66	19	16

S2_43	0.54	0.64	1535.351	7	-16.25	19	3
S2_44	0.63	0.63	1424.219	7	-14.36	18	9
S2_45	0.55	0.65	1442.057	11	-19.85	18	4
S2_46	0.76	0.68	1659.077	13	-15.23	19	9
S2_47	0.71	0.67	2016.393	11	-14.45	18	2
S2_48	0.56	0.64	1657.577	9	-18.21	17	5
S2_49	0.58	0.65	1313.481	15	-19.39	17	6
S2_50	0.57	0.64	1505.654	15	-17.11	18	11
S2_51	0.56	0.64	1959.426	15	-14.92	18	6
S2_52	0.53	0.64	1817.85	15	-18.38	17	7
S2_53	0.52	0.64	1882.13	7	-13.37	17	9
S2_54	0.55	0.65	1499.398	8	-10.21	19	4
S2_55	0.55	0.64	1959.964	12	-15	18	2
S2_56	0.69	0.67	1477.338	7	-12.07	17	20
S2_57	0.57	0.65	1958.697	9	-19.44	18	18
S2_58	0.55	0.64	1487.825	12	-15.88	17	9
S2_59	0.57	0.65	1690.861	7	-17.06	17	14
S2_60	0.54	0.64	1854.632	13	-15.83	19	19
S2_61	0.52	0.64	1966.438	12	-18.49	18	18
S2_62	0.51	0.63	1781.512	12	-16.41	17	14
S2_63	0.57	0.65	1418.472	13	-17.18	17	11
S2_64	0.56	0.65	1467.64	13	-15.26	17	11
S2_65	0.57	0.64	1670.308	15	-19.15	19	14
S2_66	0.55	0.65	1397.543	9	-9.93	18	8
S2_67	0.77	0.68	2235.785	14	-18.25	19	8
S2_68	0.53	0.64	1574.226	11	-19.87	18	14
S2_69	0.55	0.65	1633.926	12	-13.82	18	9
S2_70	0.57	0.65	1763.877	7	-18.45	18	19
S2_71	0.55	0.65	1503.103	14	-10.03	18	17
S2_72	0.53	0.65	1311.414	12	-15.65	17	3
S2_73	0.55	0.64	1865.513	8	-16.22	19	16
S2_74	0.53	0.64	1723.215	11	-15.83	19	8
S2_75	0.56	0.64	1975.551	12	-15.13	17	12
S2_76	0.59	0.65	2052.199	15	-19.28	17	14
S2_77	0.53	0.64	2131.635	12	-11.23	18	9
S2_78	0.56	0.64	1409.411	12	-12.95	19	2
S2_79	0.52	0.64	1953.141	14	-13.97	19	20
S2_80	0.53	0.64	1722.173	12	-13.46	17	5
S2_81	0.56	0.64	1740.245	10	-16.82	18	5
S2_82	0.55	0.64	1571.556	10	-18.83	17	17
S2_83	0.53	0.64	1581.117	15	-12.2	19	20
S2_84	0.54	0.64	2102.818	15	-18.46	18	20
S2_85	0.54	0.65	1878.91	11	-18.22	17	4
S2_86	0.54	0.64	1474.2	12	-12.96	17	6
S2_87	0.54	0.64	2141.376	15	-14.59	17	6
S2_88	0.57	0.64	1732.069	10	-11.29	19	9

S2_89	0.52	0.63	2140.28	10	-18.51	17	11
S2_90	0.55	0.64	2052.395	15	-16.85	17	6

A total of 90 independent AlphaFold3 predictions were performed and shown. ipTM, interface predicted TM-score (0–1, higher indicates higher confidence). pTM, predicted TM-score. BSA ($\text{\AA}^2$), buried surface area in square angstroms. For Hydrogen bond analysis, effective bonds denote the number of interfacial hydrogen bonds, and average energy represents the average interaction energy (in kcal/mol). mimic with IL6 denotes the number of interfacial hydrogen bonds that recapitulate the native binding interfaces of IL-6. For Hydrophobic pairing, effective bonds indicate the number of interfacial hydrophobic contacts.

Table 3. 90 repeated AlphaFold3 predictions of the S3 virtual complex

	ipTM	pTM	BSA ($\text{\AA}^2$)	Hydrogen bond analysis		Hydrophobic pairing	
				effective bonds	average energy	mimic with IL6	effective bonds
S3_1	0.56	0.65	2364.587	11	-14.33	17	18
S3_2	0.52	0.64	1530.069	8	-13.37	17	12
S3_3	0.53	0.64	1586.203	12	-17.36	16	10
S3_4	0.56	0.65	2029.343	19	-24.93	17	16
S3_5	0.53	0.65	2024.159	15	-20.29	16	11
S3_6	0.52	0.64	2305.656	13	-15.36	15	24
S3_7	0.55	0.65	1646.323	11	-12.43	16	7
S3_8	0.57	0.65	1423.955	11	-13.38	16	4
S3_9	0.56	0.65	1659.785	11	-12.66	16	9
S3_10	0.52	0.64	1704.409	12	-16.73	17	15
S3_11	0.55	0.64	1432.347	18	-20.63	15	11
S3_12	0.54	0.65	1464.516	9	-22.41	17	14
S3_13	0.54	0.65	1550.833	13	-14.46	15	15
S3_14	0.53	0.64	1919.658	8	-24.35	16	10
S3_15	0.53	0.64	2266.437	8	-24.73	17	6
S3_16	0.57	0.65	1692.803	11	-24.21	17	23
S3_17	0.54	0.65	1632.586	13	-20.77	17	23
S3_18	0.53	0.64	1830.883	8	-22.38	15	5
S3_19	0.53	0.65	1431.469	8	-18.64	15	15
S3_20	0.53	0.64	1726.449	18	-17.93	17	22
S3_21	0.55	0.64	1974.366	11	-24.43	16	6
S3_22	0.57	0.65	1443.184	13	-24.08	16	23
S3_23	0.55	0.65	1858.176	12	-24.38	16	13
S3_24	0.52	0.64	1547.939	12	-15.59	16	13
S3_25	0.56	0.65	1979.198	15	-17.57	17	16
S3_26	0.53	0.64	1455.148	19	-24.46	17	5
S3_27	0.53	0.64	1902.916	13	-13.68	17	14
S3_28	0.56	0.65	1636.961	17	-18.59	17	12
S3_29	0.57	0.65	1568.054	17	-21.94	17	20
S3_30	0.53	0.64	2144.254	14	-24.54	17	4
S3_31	0.53	0.64	1792.139	9	-22.3	15	15
S3_32	0.52	0.64	1429.642	10	-20.51	17	13
S3_33	0.53	0.64	1890.874	10	-13.54	15	8
S3_34	0.55	0.65	1695.895	11	-22.17	15	20

S3_35	0.53	0.65	1740.649	8	-12.47	15	19
S3_36	0.54	0.64	1663.413	8	-16.66	15	23
S3_37	0.57	0.65	1824.085	14	-22.81	16	7
S3_38	0.57	0.65	1730.576	13	-18.79	15	4
S3_39	0.55	0.65	1581.347	10	-18.28	17	12
S3_40	0.54	0.65	1660.787	13	-13.69	15	23
S3_41	0.54	0.65	2050.673	9	-21.87	16	23
S3_42	0.53	0.64	1944.503	10	-22.39	15	17
S3_43	0.53	0.63	2004.581	11	-13.32	17	20
S3_44	0.53	0.65	2098.084	19	-13.35	17	16
S3_45	0.53	0.64	1592.321	9	-17.36	15	18
S3_46	0.58	0.65	2547.74	11	-20.49	16	8
S3_47	0.55	0.65	2075.042	19	-19.93	15	5
S3_48	0.55	0.64	1782.028	15	-24.12	15	9
S3_49	0.58	0.65	2292.741	10	-17.39	16	11
S3_50	0.53	0.64	1737.644	11	-12.57	15	19
S3_51	0.58	0.65	2452.06	13	-17.95	17	4
S3_52	0.58	0.65	1967.12	12	-13.93	17	7
S3_53	0.54	0.65	1594.367	14	-22.61	16	22
S3_54	0.54	0.65	2162.878	16	-17.37	17	24
S3_55	0.52	0.64	1630.13	10	-12.52	17	8
S3_56	0.53	0.64	1700.855	15	-15.27	17	5
S3_57	0.53	0.64	1455.658	10	-23.72	15	19
S3_58	0.53	0.65	1589.55	14	-14.92	16	12
S3_59	0.56	0.65	1916.114	17	-16.06	16	22
S3_60	0.54	0.65	1780.142	13	-24.53	15	21
S3_61	0.54	0.65	1674.459	17	-14.71	16	20
S3_62	0.53	0.64	1797.117	18	-21.31	17	5
S3_63	0.53	0.64	1624.347	17	-12.47	16	4
S3_64	0.55	0.64	1881.191	18	-20.08	17	8
S3_65	0.6	0.66	2050.174	14	-24.85	16	18
S3_66	0.52	0.64	2083.177	13	-15.81	17	18
S3_67	0.53	0.64	1887.649	10	-19.91	17	5
S3_68	0.53	0.64	1526.603	16	-23.14	15	9
S3_69	0.55	0.65	1597.863	15	-23.21	17	24
S3_70	0.55	0.65	1912.613	12	-13.38	16	13
S3_71	0.56	0.65	2068.513	18	-24.84	17	16
S3_72	0.57	0.66	1694.703	12	-20.96	16	17
S3_73	0.53	0.64	1517.825	19	-18.19	15	18
S3_74	0.54	0.65	2038.091	14	-23.08	16	17
S3_75	0.54	0.65	1792.975	9	-14.72	15	4
S3_76	0.52	0.64	1625.174	9	-19.03	15	18
S3_77	0.53	0.64	2079.439	17	-16.46	15	23
S3_78	0.53	0.64	1550.942	16	-24.92	17	8
S3_79	0.54	0.65	1702.658	15	-17.1	15	10
S3_80	0.53	0.64	1685.096	10	-23.99	15	20

S3_81	0.53	0.64	1979.595	10	-20.16	16	24
S3_82	0.53	0.64	1663.904	11	-18.96	17	15
S3_83	0.53	0.64	2028.658	17	-22.59	17	20
S3_84	0.53	0.64	1586.073	13	-16.55	15	5
S3_85	0.5	0.62	1714.102	19	-12.8	16	19
S3_86	0.54	0.64	2144.583	16	-22.4	15	13
S3_87	0.53	0.64	1576.106	14	-17.08	16	22
S3_88	0.58	0.65	1648.126	17	-21.81	15	4
S3_89	0.53	0.64	1453.153	19	-18.59	16	8
S3_90	0.56	0.65	1609.144	9	-14.56	16	19

A total of 90 independent AlphaFold3 predictions were performed and shown. ipTM, interface predicted TM-score (0–1, higher indicates higher confidence). pTM, predicted TM-score. BSA ($\text{\AA}^2$), buried surface area in square angstroms. For Hydrogen bond analysis, effective bonds denote the number of interfacial hydrogen bonds, and average energy represents the average interaction energy (in kcal/mol). mimic with IL6 denotes the number of interfacial hydrogen bonds that recapitulate the native binding interfaces of IL-6. For Hydrophobic pairing, effective bonds indicate the number of interfacial hydrophobic contacts.

Table 4. Interface hotspot residues (Z-score > 2.00) identified by contact-based energetic scoring in the second round of evolution

Model	Residue	Chain	Score	Z-score
S1	TYR13	IL6	450	3.25
S1	ARG6	IL6	351	2.35
S1	LEU1	IL6	342	2.27
S1	TRP169	sIL6R	664.5	5.12
S1	TYR50	mimic	441	3.16
S1	TYR25	mimic	403	2.82
S2	TYR13	IL6	438.5	3.16
S2	ARG6	IL6	354	2.39
S2	LEU1	IL6	346	2.32
S2	TRP169	sIL6R	1880.5	8.21
S2	LEU172	sIL6R	841	3.3
S2	TYR50	mimic	431	3.09
S2	TYR25	mimic	382	2.64
S3	TYR13	IL6	437.5	3.14
S3	LEU1	IL6	358.5	2.43
S3	ARG6	IL6	349.5	2.34
S3	ARG118	sIL6R	260.5	4.43
S3	ARG115	sIL6R	205.5	3.35
S3	TYR50	mimic	440	3.16
S3	TYR25	mimic	387	2.68
S1	TYR13	IL6	450	3.25
S1	ARG6	IL6	351	2.35

List of predicted interface hotspot residues across the evolved models (S1, S2, and S3). Hotspots were defined as residues with a Z-score > 2.00 based on contact-based energetic scoring. Model, identifier of the evolved design model. Residue, amino acid type and sequence position of the identified hotspot. Chain, the binding interface context where the hotspot is located, where IL6 and sIL6R denote residues at the native IL-6 and sIL-6R binding interfaces, respectively, and mimic denotes residues on the binder interface that mimics the native binding mode. Score, the

contact-based energetic score of the residue. Z-score, the standard score indicating the statistical significance of the energetic score relative to the background distribution.

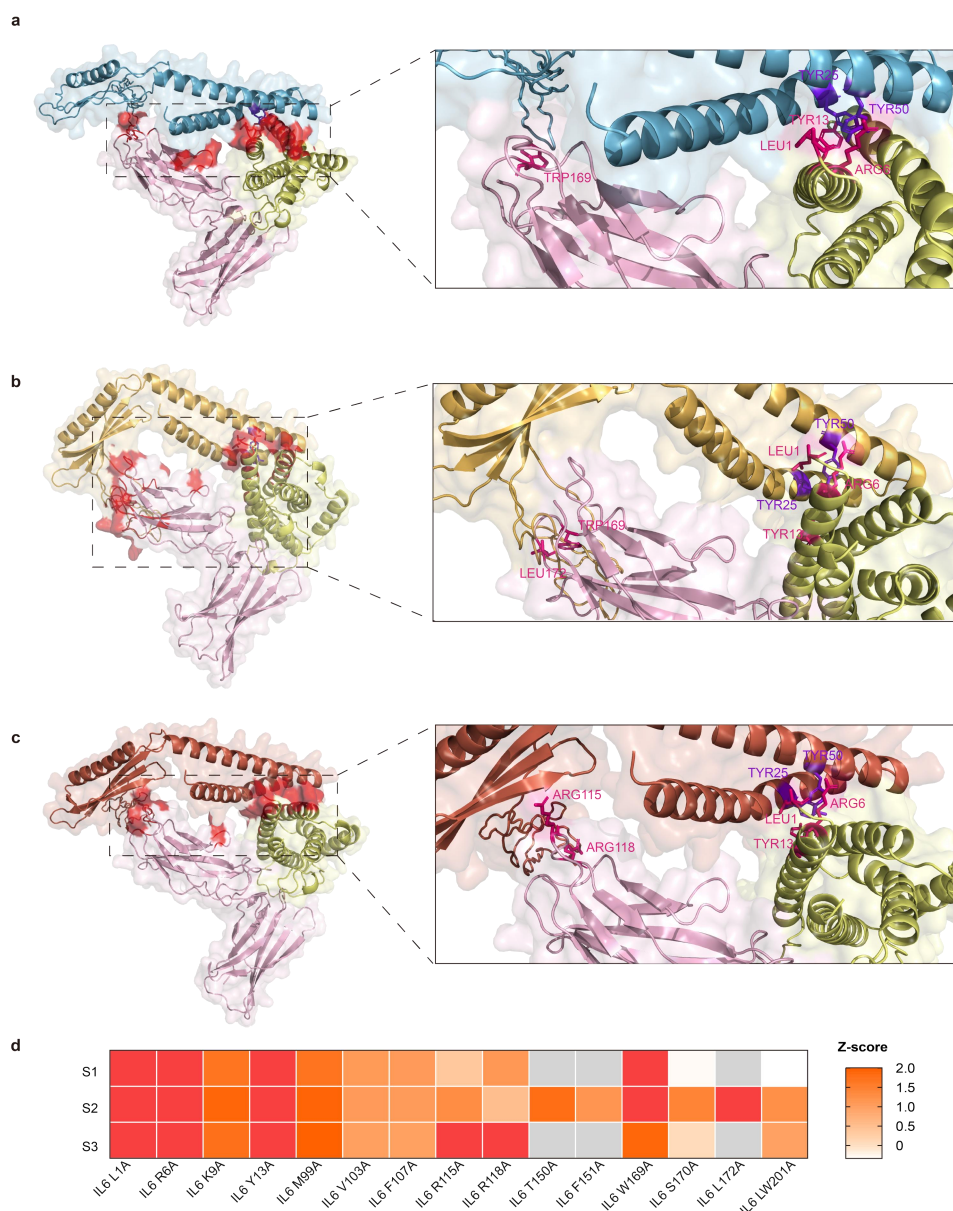


Figure 1. In silico construction of second-round evolution candidates. a-c, Ninety repeated predictions of S1, S2, and S3 were performed using AlphaFold3. The five predictions with the highest ipTM scores were selected as high-confidence models, and the averaged virtual complex structures derived from their superposition (left), together with the hot-spot residues identified by alanine scanning (Z-score > 2) (right), are shown. S1, S2, and S3 are shown in cyan, gold, and salmon, respectively; IL-6 and sIL-6R α are shown in yellow and pink, respectively. Hot-spot residues on the IL-6/sIL-6R α complex are highlighted in magenta, and those on the candidates are highlighted in purple. The binding interface is highlighted in red in the left panels. d, Heatmap of Z-scores for hot-spot residues on the IL-6/sIL-6R complex side. The vertical axis represents the three candidates, and the horizontal axis corresponds to alanine mutations of the hot-spot residues. The white-to-orange gradient indicates increasing Z-scores, red denotes hot-spot residues with Z-scores > 2, and grey indicates residues absent from the binding interface of the corresponding complex.

7. AI Use Statement:

AI-assisted structure prediction and interface analysis were performed. Computational predictions were distinguished from experimental validation.